The Origin of Species Revisited: A Necessity-Based Model with Discontinuous Boundary Conditions

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Abstract

Classical Darwinian evolution explains adaptive variation under slowly varying environmental conditions through stochastic mutation and natural selection. However, multiple empirical events — including the Cambrian Explosion, extremophile emergence, radiation resistance, and abrupt phenotypic stabilization — reveal systematic failures of gradualism under discontinuous boundary shifts.

This paper proposes a revised framework for the origin of species based on the **Theorem of Axiomatic Necessity (TNA)**, introducing a minimum coherence constraint F_{min} that governs viability under changing conditions. We distinguish *evolution* as a local optimization mechanism from *speciation* as a necessity-driven reconfiguration triggered by boundary discontinuities. The result is a dual-regime theory in which Darwinian evolution survives as a limiting case, while the origin of species is redefined as a structural phase transition rather than a stochastic search process.

1. Introduction: Why “Origin of Species” Must Be Rewritten

Darwin’s *On the Origin of Species* implicitly assumes that the same mechanism responsible for variation also explains emergence. This conflation has remained largely unchallenged due to the empirical success of microevolutionary models.

However, the **origin** of species is not equivalent to their **subsequent evolution**.

This paper argues that:

- Darwinian evolution is valid **only locally**
- Speciation requires a **distinct mechanism**
- Abrupt environmental changes invalidate stochastic gradualism
- A minimum coherence constraint F_{min} governs survivability

2. Definitions (Critical Clarification)

Definition 1 — Evolution

Evolution is a continuous, stochastic optimization process acting on an already viable architecture under approximately stable boundary conditions.

It operates by:

- small mutations
- selection differentials
- local fitness gradients

Evolution **cannot generate new global architectures**.

Definition 2 — Origin of Species

The **origin of a species** is the emergence of a new coherent biological architecture capable of persistence under a given set of boundary conditions.

This is a **global structural event**, not a local optimization.

Definition 3 — Boundary Conditions

Boundary conditions are external constraints (chemical, energetic, ecological, radiative, climatic) that define the viability space of biological systems.

They may vary:

- continuously (slow drift)
- discontinuously (phase shifts)

Definition 4 — Minimum Coherence Constraint ($F_{\min}$)

$F_{\min}$ is the minimal structural coherence required for a biological system to persist.

If system coherence $C < F_{\min}$, the system collapses regardless of adaptive potential.

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3. The Failure of Darwinian Mechanisms Under Discontinuity

Darwinian evolution assumes:

- continuity of fitness landscapes
- gradual accessibility of intermediate forms
- sufficient time for stochastic search

Under abrupt boundary shifts:

- most mutations are lethal
- intermediate forms are nonviable
- search space explodes combinatorially

Formally, the probability of stochastic convergence under discontinuity tends to zero:

$$\lim_{\Delta B \to \text{discontinuous}} P_{\text{Darwin}} = 0$$

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4. The TNA Speciation Mechanism

Core Principle

> ***Speciation is not the result of accumulated variation, but of enforced coherence under necessity.***

When boundary conditions shift discontinuously:

1. Existing architectures lose coherence
2. The system approaches collapse
3. Only configurations satisfying $F_{\min}$ persist
4. A new species appears as a **crystallized solution**, not a searched one

This is a **phase transition**, not an evolutionary trajectory.

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5. Dual-Regime Model of Life Dynamics

REGIME	BOUNDARY CONDITIONS	MECHANISM	OUTCOME
Stable	Continuous	Darwinian evolution	Optimization
Unstable	Discontinuous	TNA reconfiguration	Speciation or extinction

Evolution acts **after** speciation, never as its cause.

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6. Reinterpreting Canonical Events

Cambrian Explosion

Not a burst of successful mutations, but a **global boundary shift** forcing architectural reconfiguration.

Extremophiles

Not gradual adaptation, but survival of pre-compatible architectures under new constraints.

Radiation Resistance (Chernobyl, Bugorski)

Immediate coherence preservation under novel boundary regimes.

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7. Why “Extended Evolutionary Synthesis” Is Insufficient

The Extended Synthesis:

- adds epigenetics
- includes niche construction
- introduces plasticity

But it **retains continuity assumptions** and does not address:

- computability limits
- architectural jumps
- probability collapse under stress

Thus, it extends evolution but **still cannot explain origin**.

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8. Information-Theoretic Limit

A system cannot compute new boundary conditions from internal information alone.

Therefore:

- speciation cannot be algorithmic
- adaptation cannot precede coherence
- architecture precedes optimization

This establishes a hard limit on evolutionary explanations.

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9. Conclusion: A New Origin of Species

Darwin explained how species **change**.
He did not explain how species **begin**.

The origin of species is governed by:

- necessity, not chance
- coherence, not optimization
- discontinuity, not accumulation

Evolution survives as a **local approximation**.
Speciation is a **structural response to boundary necessity**.

This distinction resolves long-standing paradoxes without invoking metaphysics or probabilistic miracles.

Final Statement

> ***Species do not originate by searching possibility space.***
> ***They originate when reality withdraws all other options.***